WASP-2b

R-1393

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-2_220718` · created 2026-08-01T03:17:31+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459778.8306 +/- 0.0072 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.047
Transit depth (Rp/Rs)^2	2.55 +/- 1.49 [%]
Orbital Inclination (inc)	83.42 +/- 2.92
Airmass coefficient 1 (a1)	0.791 +/- 0.036
Airmass coefficient 2 (a2)	0.21 +/- 0.28
Scatter in the residuals of the lightcurve fit is	4.63 %
Best Comparison Star	#1 - [508, 358]
Optimal Aperture	3.06
Optimal Annulus	9.68
Transit Duration (day)	0.05 +/- 0.037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.16 ± 0.047 vs 0.1326 (deviation 0.58σ)
Duration fit vs published	0.05 d vs 0.0737 d (deviation 0.64σ)
Mid-time O–C	-13.9 min
Depth SNR	3.4
Residual scatter	4.63 %

Light curve

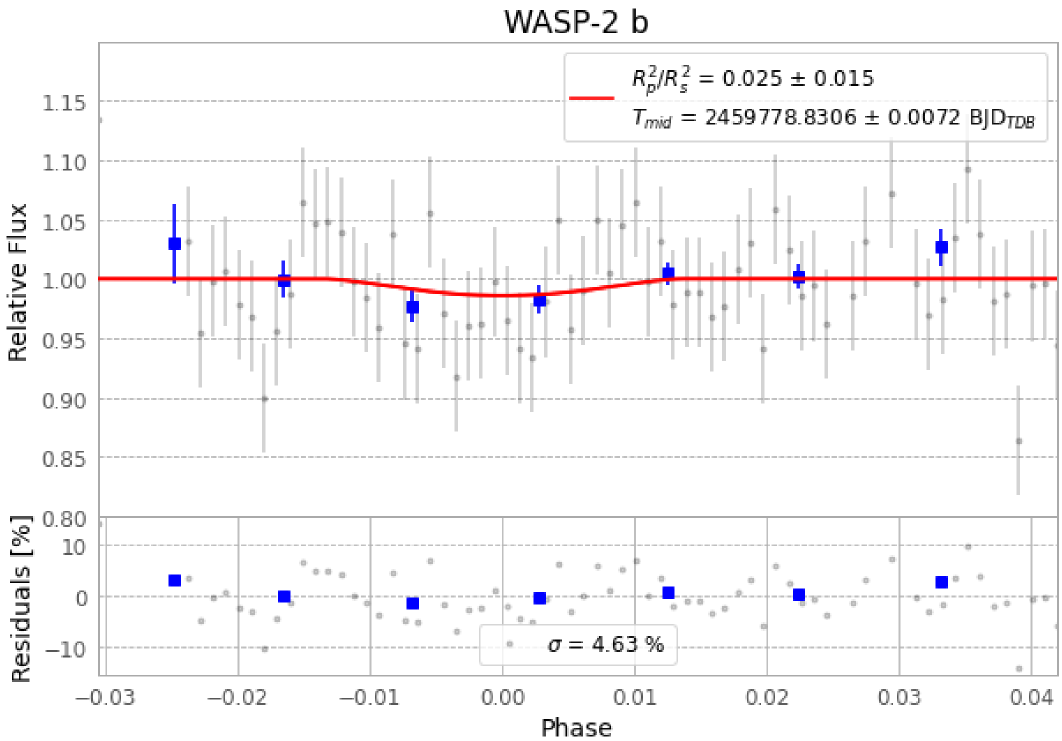

Figure 1 — FinalLightCurve_WASP-2 b_2022-07-17.png (SHA-256 15a093ad8afb45ef…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [467, 241], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 cede56485c587b16…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

76 FITS frames (50 MB), `WASP-2220718061213.FITS` → `WASP-2220718095713.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-220718.log (e2b2643f4a21…), FinalLightCurve_WASP-2 b_2022-07-17.png (15a093ad8afb…), FinalLightCurve_WASP-2 b_2022-07-17.csv (dae351d2f5f9…)

Compute resources

Wall time 445.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)